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# Hinterland operations of sea ports do matter: Dry port usage effects on transportation costs and $CO_2$ emissions

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ABSTRACT

Decreasing carbon dioxide ($CO_2$) emissions is one of the most important tasks for the society in the 21st century. One possibility to decrease emissions originating from transportation is to utilize more rails instead of relying simply on road transportation. In the dry port concept an inland intermodal terminal is connected to a sea port using railways. This study analyzes impacts of dry ports in a Finnish context. We compare two different configurations: In the first one shippers drive directly to a sea port, while in the second one they use dry ports. The systems are evaluated by using discrete-event simulation. In the systems we are interested in two issues: (1) Level of $CO_2$ emissions, and (2) Costs to transport the goods in different configurations. We use different scenarios for future energy prices and estimate both the costs and $CO_2$ emission development in these scenarios. We also compare the results to a situation, where emissions are minimized instead of costs. Implications on larger scale are also discussed, for example in the Baltic Sea and North Sea area, where strict sulfur emission restrictions are seen to harm sea transport and increase concentration on small number of sea ports.

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## 1. Introduction

European Union (EU) is paying large attention to decreasing environmental impacts originating from transportation, which is one of the most emitting sectors (Aronsson and Brodin, 2006; European Commission, 2001). Furthermore, transportation is the only sector that has increased its emissions year by year (European Commission, 2009; UIC, 2009). Its share from overall greenhouse gas (GHG) volume is one fourth and is still growing (see Stead, 2006). Result of numerous research works is that environmental impacts of transportation can be decreased by decreasing the amount of road transport and substituting it with some other transport mode, which is environmentally friendlier than road transport (see e.g. LIPASTO, 2009; Maibach et al., 2008; Van Wee et al., 2005).

The IMO Marpol 73/78 Annex VI regulations aim to reduce nitrogen oxide (NOx) emissions and prevent sulfur oxide (SOx) and particular matter emissions from sea transportation. SOx amounts have to be decreased radically in Sulphur Emission Control Area (SECA), which includes North Sea and Baltic Sea. SOx amount of fuel cannot exceed 0.10% after January 2015 (already implemented with 1.0% max. sulfur content in Baltic Sea). It should be noted that in IMO's action plan whole world will implement low sulfur regulation of 0.5% from year 2020 onwards. Interestingly IMO has not stated any further actions to achieve the level, where e.g. Baltic Sea is demanded to be already from year 2015 onwards. It is possible that short sea shipping will decrease its modal share due to stricter SOx regulations e.g. in case of Finland

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and Baltic States. Reason is 30–40% higher costs of sea transport e.g. between Finland and Germany (Kalli et al., 2009; in general Delhaye et al., 2010). Substitute for decreased sea shipping will most probably be hinterland transportation in form of road or rail transport. Furthermore, sulfur regulations will likely increase concentration on fewer number of sea ports i.e. most sea vessels will use southern sea ports in Finland to minimize time spent in SECA. This will lead in longer hinterland transportation routes for road and rail transport. (Det Norske Veritas, 2009; ECSA, 2010; Henttu and Hilmola, 2011; Swedish Maritime Association, 2009)

One way of increasing the usage of rail transport is to implement dry ports. Dry port concept is part of intermodal transportation, which develops sea port connected inland transportation. In the dry port concept transportation between sea port and inland intermodal terminal is accomplished mainly by rail transport. In addition, some or all services usually conducted in sea port are transferred to dry ports. One of the main benefits of the dry port concept is the decreased environmental harms caused. Total $CO_2$ emissions originating from transportation decrease as road transport is substituted with rail transport and rail transport emits less $CO_2$ emissions per ton-kilometer than road transport (Janic and Vleugel, 2012; LIPASTO, 2009; Maibach et al., 2008; Van Wee et al., 2005). In addition, economies of scale are possible to achieve by using rails, which has a larger capacity than road transport. It is also possible to decrease the amount of accidents, level of congestion and noise (Henttu et al., 2010; Korovyakovsky and Panova, 2011; Notteboom and Rodrigue, 2005; Roso, 2007; Roso et al., 2009; Roso and Lumsden, 2010).

In order for a dry port to be successful, customers need to be able to operate with lower (or at least equal) costs compared to a direct road connection (Cullinane and Wilmsmeier, 2011), but at the same time the dry port should be able to generate enough volume (Rodrigue and Notteboom, 2012). As the cost structure for intermodal freight transport systems are highly dependent on the cost of rail and truck transport (Kim and Van Wee, 2011), it is also important to understand the feasibility of the dry port systems in future conditions. Road transportation is currently a very competitive transport mode, since it is flexible with an attractive price level. But the price level can change as the price of oil is predicted to continue to rise in the future.

One important aspect in dry ports is the regional perspective. According to Rodrigue and Notteboom (2012) the interaction between dry ports and their regional markets is a fundamental issue. It is even difficult to estimate a break-even distance for an intermodal freight transport system as many different issues impact this distance (Kim and Van Wee, 2011). Van den Berg and De Langen (2011) also argue that concentrating only in road transportation restricts sea ports chances in gaining additional market share. As such it is important to correctly estimate the appropriate locations where a dry port can be located. Wilmsmeier et al. (2011) have also identified spatial directional development of intermodal terminals as an important research agenda. At the same time, there is a potential for overinvestment as many locations would like to play a crucial role in global value chains (Rodrigue and Notteboom, 2012). According to Bergqvist and Wilmsmeier (2009) effective and coordinative governance and actions are needed both at a national level as well as on an EU level in order to cope with the problem of overinvestments. Government policies are needed to enable investments in ideal locations to create a good transport infrastructure.

The aim of this research is to find out the feasibility of dry ports in Finland in different scenarios. As dry ports are part of larger systems (Rodrigue et al., 2010; Roso et al., 2009) and they are influenced by many factors (Flämig and Hesse, 2011), it is also important to study dry ports more holistically in larger geographical areas. The first main research question of this study is: "How can the locations of potential dry ports be estimated on a regional scale?" We assume that the operators are interested only in lowering their own costs, but we also study the environmental impact of the potential dry port structures. The research question can also be divided to smaller research questions: "How can the stability of demand for dry ports be estimated?" and "How do potential dry ports interact in larger logistical systems?". In addition to these we analyze the volumes for these potential dry ports. The second main research question is: "How much can the emissions of transportation be lowered by utilizing dry ports?" This is an important issue due to the need to decrease GHGs. Governmental decision-makers need to understand the environmental impact of dry port systems.

As dry port systems tend to consist of many different actors (Rodrigue and Notteboom, 2012; Flämig and Hesse, 2011; Haralambides and Gujar, 2011), it is vital to achieve buy-in from all of these individual organizations. Several studies regarding intermodal terminal locations using analytical models have been presented in the literature (see e.g. Henttu and Hilmola, 2011). However, simulation is generally regarded as a good way to achieve buy-in for complex issues (Robinson, 2004) and provides a way to enforce learning from analytical models (Banks et al., 2005). In this research the main research method is discrete-event simulation modelling, where the model simulates the Finnish transportation network. The transportation flows in the model are based on population and we provide an overview of the structure of the model. The model can be replicated to other regional systems as well to analyze potential dry port systems in other geographical areas. We use the simulation model to study different scenarios in Finland and test the networks with different cost structures. The results of the model can be used to estimate the best locations for dry ports as well as other inland terminals.

Structure of this paper is as follows: Section 2 provides a literature review concerning dry port concept, oil price, and future outlook of oil price development. Section 3 clarifies research methodology of this paper and introduces the simulation model. Thereafter, in Section 4 we analyze the results of the simulation experiment. Discussion regarding the role of $CO_2$ emissions on overall simulated system economy is given in Section 5. And in final Section 6 we conclude our work, and provide future avenues in this endeavor.

## 2. Literature review

### 2.1. Dry port concept

Traditional sea ports have pressure to improve their competitiveness in comparison to other sea ports to maintain or increase their market share. With dry port concept sea ports can increase their service levels, capacities and storage areas in inland located intermodal terminals, which are connected to sea ports with rail link. Due to increased need of effective hinterland transport, dry ports have been studied widely during the recent years. Table 1 summarizes these contemporary research findings of dry ports. (See e.g. Bergqvist and Egels-Zandén, 2012; Cullinane and Wilmsmeier, 2011; Flämig and Hesse, 2011; Hanaoka and Regmi, 2011; Haralambides and Gujar, 2011; Henttu and Hilmola, 2011; Ka, 2011; Korovyakovsky and Panova, 2011; Notteboom and Rodrigue, 2005; Rodrigue et al., 2010; Roso, 2008; Roso et al., 2009).

Roles of different stakeholders in dry port implemented transportation system have been studied and discussed by many authors. There are usually numerous amount of stakeholders involved in transportation network, which includes dry ports (Rodrigue and Notteboom, 2012; Flämig and Hesse, 2011; Haralambides and Gujar, 2011). The amount of stakeholders will increase, while sea port is expanding in hinterland area. This will also increase the role of communication between local and global stakeholders (Bergqvist and Egels-Zandén, 2012). Furthermore, public–private-partnership plays a significant role in dry port investments and maintenance operations due to the fact that monetary investments tend to be significant (Haralambides and Gujar, 2011). Bergqvist and Wilmsmeier (2009) point out that not only national coordination, but also EU-wide cooperation is important. Increased amount of stakeholders in the system and also the hinterland expansion considerably increase the importance of the quality of hinterland strategy by both the sea port and the dry port (Van den Berg and De Langen, 2011).

Possibilities to decrease transportation costs with dry ports is widely researched subject (Chang et al., 2008; Choong et al., 2002; Cullinane and Wilmsmeier, 2011; Flämig and Hesse, 2011; Henttu and Hilmola, 2011; Henttu et al., 2011). It is important that implementing dry port will lead to decreased transportation costs, if compared to current transportation costs (Cullinane and Wilmsmeier, 2011). Estimating transportation costs from dry port implemented transportation network is more complicated than estimating costs from unimodal road transport due to increased amount of variables, which have influence in total costs (Kim and Van Wee, 2011). Without transportation cost benefits, it usually is not beneficial to invest in creation of a dry port. There are also environmental benefits to be achieved by utilizing dry port (Bergqvist and Egels-Zandén, 2012; Hanaoka and Regmi, 2011; Henttu et al., 2011; Liao et al., 2009), but these benefits are usually not enough to justify creation of a dry port without considerable internal transportation cost benefits.

Many research studies regarding dry ports are focusing in how to choose geographical location for inland intermodal terminals. Rahimi et al. (2008) developed a model, which can be used to find locations for inland ports in a way that daily

**Table 1**
List of reviewed dry port articles.

| Authors | Public–private-partnership (Stakeholder analysis) | Costs of transportation | Throughput | Regionalization and regional case studies | Container management | Environmental benefits |
|---|---|---|---|---|---|---|
| Bergqvist and Egels-Zandén (2012) | x | x | | | | x |
| Chang et al. (2008) | | x | | | x | |
| Choong et al. (2002) | | x | | x | x | |
| Cullinane and Wilmsmeier (2011) | | x | x | | | |
| Flämig and Hesse (2011) | x | x | | x | | |
| Hanaoka and Regmi (2011) | x | | | x | | x |
| Haralambides and Gujar (2011) | x | x | | x | | |
| Henttu and Hilmola (2011) | | x | | x | | x |
| Henttu et al. (2011) | | x | | x | | x |
| Janic and Vleugel (2012) | | | | x | | x |
| Ka (2011) | | x | | x | | |
| Kim and van Wee (2011) | | x | | x | | |
| Korovyakovsky and Panova (2011) | | | x | x | x | |
| Liao et al. (2009) | | | | x | | x |
| Monios (2011) | x | | | x | | |
| Notteboom (2010) | | | x | x | x | |
| Notteboom and Rodrigue (2005) | | | x | x | | |
| Rahimi et al. (2008) | | | x | x | | |
| Rodrigue et al. (2010) | x | | x | x | | |
| Rodrigue and Notteboom (2010) | | | | x | x | |
| Rodrigue and Notteboom (2012) | x | | | x | | |
| Roso (2008) | x | | | x | x | |
| Roso et al. (2009) | x | | | x | | |
| Roso and Lumsden (2010) | x | | | x | | |
| Shintani et al. (2010) | | x | | | x | |
| Song and Dong (2011) | | x | | | x | |
| Van den Berg and De Langen (2011) | x | | | x | x | |

vehicle-miles are minimized. As more inland ports are added to the model, the vehicle-miles are reduced considerably, which also decreases congestion and amount of air pollution. Furthermore, Roso (2008) focused in researching different factors, which are crucial, when implementing dry ports. Ka (2011) developed an application, which utilized fuzzy analytical hierarchy process (AHP) in seeking optimal dry port locations in China. In their study, Flämig and Hesse (2011) found out that placing dry ports could increase the amount of planning conflicts, which differentiate from conflicts arising, when expanding the actual sea port. New conflicts are usually based on demand of land area and infrastructure development for the dry port.

In regionalization the sea port basically expands its actions in its hinterland area as recognizing the importance of hinterland transport. By regionalization sea port aims to improve services it is providing to customers and also to reduce logistics costs (Notteboom and Rodrigue, 2005; Rodrigue and Notteboom, 2010). There are many local and global case studies, which have researched current dry port and regionalization development in chosen geographical locations: In his study, Notteboom (2010) has focused on European sea ports. The study reveals European container port system and main hinterland corridors. Haralambides and Gujar (2011) have focused in the development of dry port sector in India. Korovyakovsky and Panova (2011) studied current development of Russian dry ports, whereas Roso and Lumsden (2010) have studied main dry ports situated in different continents. Role of development of Spanish inland terminal has also been studied by Monios (2011). Furthermore, Hanaoka and Regmi (2011) have made an overview of the main dry ports in Asia. There is also a comparison between European and North American intermodal rail systems accomplished by Rodrigue and Notteboom (2012). They found out that there are many similarities and differences due to regional and local governance and regulations regarding dry ports and rail connections.

Since dry port concept is part of intermodal transportation, management of containers plays important role. Empty container management has been studied by Choong et al. (2002). They have suggested that empty containers need to be taken into careful consideration, when managing container flows. If only loaded containers are paid full attention, but no attention is given to empty containers, it is impossible to make profits from transportation business. Choong et al. (2002) suggest that longer planning horizon encourages using more inexpensive and slower transport modes as e.g. barge, but shorter planning horizon demands faster and more expensive deliveries. Furthermore, amount and location of container storage areas have large effect on transport mode choice. Chang et al. (2008) implemented heuristic approach to research empty container substitution problem. Their approach allowed cost reductions of empty container management of 4–47%. In addition, Song and Dong (2011) used heuristic algorithm to minimize empty container repositioning costs. Empty container management problem has also been researched by using foldable containers instead of traditional ones (Shintani et al., 2010). Results show that repositioning costs can be decreased by using foldable containers.

Environmental benefits can be achieved by intermodal transport. In their study, Liao et al. (2009) compared $CO_2$ emissions among two different scenarios: In the first, truck-only transportation is used for hinterland transportation, and in the second intermodal coastal shipping is used for long haulage, and road transport for final transportation leg. Results show that $CO_2$ emission amounts can be decreased considerably by replacing road transport with intermodal transport (reduction can be up to 60%). According to Hanaoka and Regmi (2011), utilization of Asian dry ports has been able to decrease congestion and vehicle emissions. In their study, Janic and Vleugel (2012) calculated that by substituting road with rail transport on a chosen Trans-European freight transport corridor externalities could be decreased by approximately 30%.

As the review points out, dry ports are a multi-faceted problem. The issue covers many different stakeholders and public–private-partnership needs to be taken into account when considering potential dry ports. For individual shippers, the main area of interest is nevertheless actual costs of transportation. As transportation costs are highly dependent on the global oil prices, it is important to understand, how the price of oil will fluctuate in the future.

### 2.2. Long term development of oil prices

For a long period of time, oil prices in the world were at extremely low levels (i.e. a couple of USDs per barrel). Economic problems, abandoning Bretton Woods currency arrangement (year 1971; Sarnoff, 1980) and eventually oil crises in the 1970s changed the level of these prices, which increased up to 30–40 USD per barrel in the early 1980s. Thereafter, for two decades prices remained at relatively low levels, 15–20 USD per barrel.

However, as free trade movement spread round the world (e.g. China joining WTO in late 2001), these new emerging economies needed to fuel their energy needs with 'old' fossil fuels such as coal (Lin and Liu, 2010) and oil (Tao and Li, 2007). Actually these two main fossil fuels got considerable support for their usage from emerging Asian markets, and particularly from China (actually coal consumption has now more than doubled in Asia Pacific Region within previous decade, and China nowadays accounts for nearly half from the entire world consumption). Increase in the e.g. oil demand created oil bubble, which burst in year 2008 (Fig. 1 and Appendix A; please do note that oil price is 200 days moving average).

The enormous increase in oil demand created overheating in prices, and bubble burst rather rapidly: Oil price per barrel (European Brent) was having valuation of 140 USD per barrel in July 2008, but slumped down within six months to the level of 35 USD! However, this sudden decline was caused by realization of systemic crisis in the world economy, and central banks reacted strongly on this situation by providing massive amount of liquidity to the markets. This in turn has helped critical commodities to inflate their values as banks and investment community has been on the search of safety for provided liquidity (also in the case of oil, inelastic demand and limited supply play roles in the pricing, please see El-Gamal and Jaffe, 2010). It is surprising, how closely central bank credit funding (in US case QE1, QE2 and Operation Twist waves) go hand in

hand with the price of oil. End result of the extra liquidity provided is that oil prices were already in the end of year 2011 (as measured with 200 days moving average) higher than in the peak year 2008 (price peak was very sudden earlier, and high prices did not sustain that long). The robustness of this price recovery is very troublesome for transport logistics sector, especially for road transport, which is tightly tied upon oil usage.

Market price of oil is having high correlation with refined products of oil, such as diesel oil (used in trucks and partly in train traction) or heavy oil (used in ships). However, tax policies, environmental charges and refining companies operating costs and margins have occasionally higher impacts than what would oil market price changes alone indicate (e.g. shipping oil, bunker prices in year 2011; United Nations, 2012).

One major supporting factor (along with monetary policies applied) in oil pricing is its limited supply (based on both IEA, 2011 and BP, 2012 statistics indicate that oil production is slowing, especially in Middle East). This could be illustrated with long-term statistics through the difference of oil production and oil consumption in the five largest oil producers in the world as is shown in Fig. 2. Within the long observation period both USA and China have ended up in the situation, where oil imports are continuously increasing (in USA after 2005 trend has a bit changed), and these two countries could be described to be entirely dependent on foreign oil. This would be sustainable, if other countries could develop massive surpluses. However, this is not any longer the situation (supporting oil peak theory, e.g. Reynolds and Kolodziej, 2008; Maggio and Cacciola, 2009). Saudi Arabia has been struggling to increase its production volumes, while domestic consumption has considerably increased – in the end their export potential is decreasing and volumes are around 400 million tons in Fig. 2 (production minus consumption in here is export). Russia and Iran do not perform that much different in this respect; both are having problems to increase their surplus significantly (actually Russian oil export volumes in tons are not increasing based on their official statistics, mostly due to domestic direct and indirect consumption; see Central Bank of Russian Federation, 2012). If we would calculate production and consumption of these five countries together, we may identify that surplus along with them is continuously declining, and could even enter negative territory rather soon (years 2013–2014, if long-term trend continues).

Even if oil price increase has hurt (and will do so based on the scenario of this research work) national economies and especially trade balances (e.g. Sandalow, 2008), it is on the other hand very important source of governmental budgets (taxes are mostly applied on purchased fuel). Queiroz et al. (2008) illustrate that road transport, and particularly private car usage, contribute several times that of costs of road maintenance and renewal in Europe (highest gathered amount with respect of road sector costs are in Italy, France and Germany). Of course European countries are increasingly having road tolls and vign-

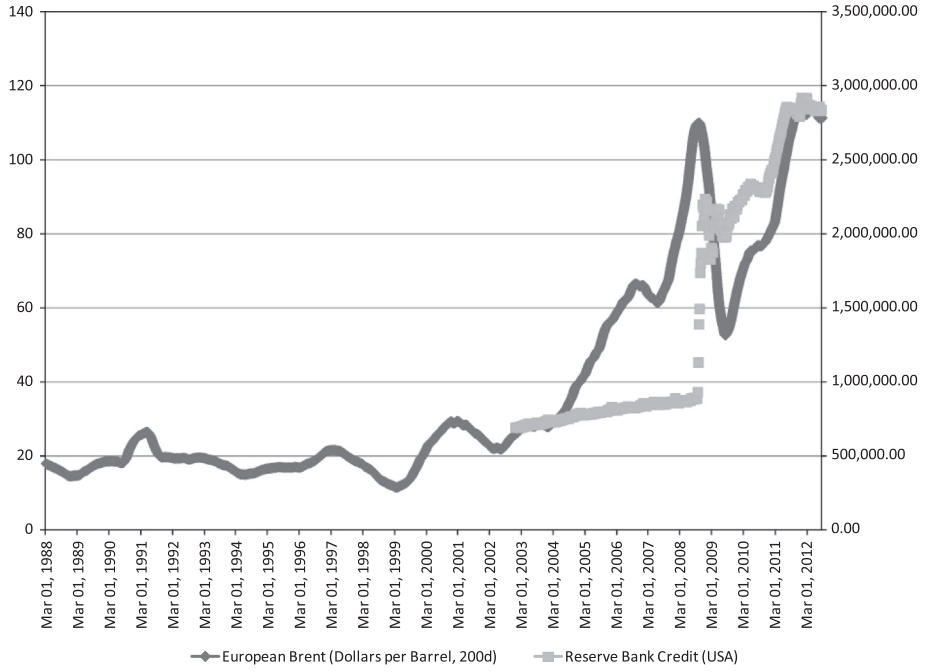



**Fig. 1.** European Brent (European Brent blend refers to oil drilled out of North Sea, North-West of Europe. It is light oil, having low sulfur content. Brent is popular oil among refineries due to its technical characteristics, but notable is that it is a bit more expensive than sweet crude (latter is used as typical blend in price reporting around the world). However, European Brent is more popular in oil supply contracts throughout the world. (Neste Oil, 2013)) spot market price (USD per barrel) 200 days moving average (dark curve, *y*-axis left-side; USD per barrel) during May 1988–August 2012, and US Federal Reserve Bank Credit (Reserve Bank Credit (see Federal Reserve, 2013) includes money loaned by central bank to operating banks, purchased debt from markets (e.g. governmental bonds), and other loans made (e.g. to troubled institutions/banks).) (gray curve, *y*-axis left-side; mill. USD) during December 2002–August 2012 (mill. USD). *Source* (data): EIA (2012) and US Feredal Reserve (2012).

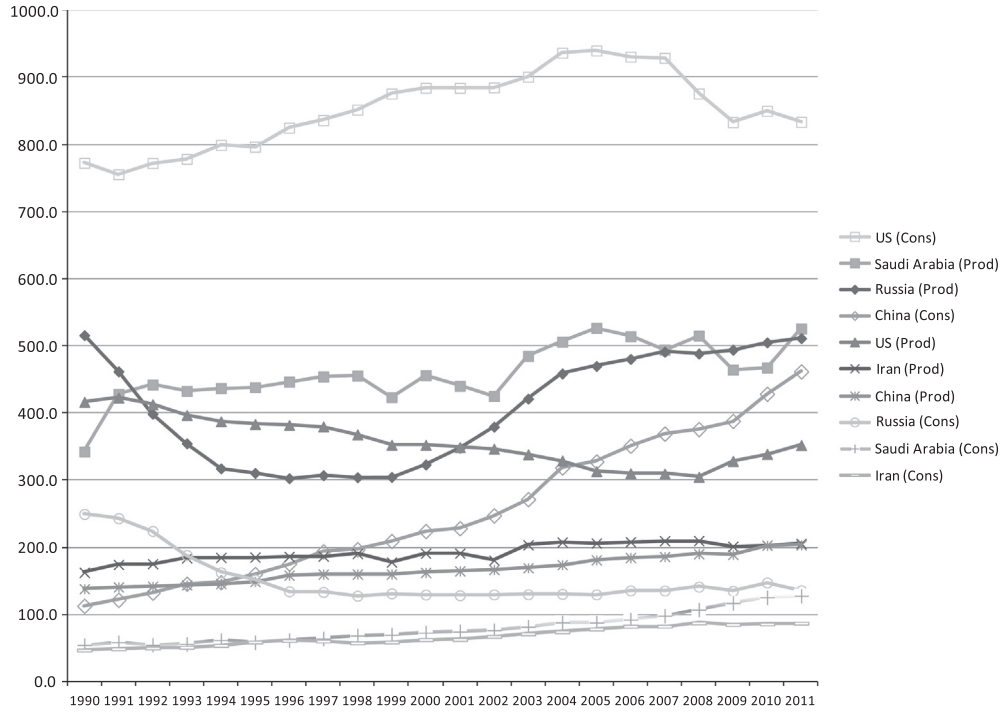



**Fig. 2.** Oil production in a country and home market consumption during period of 1990–2011 – five most important oil producers in the world analyzed (million tons). *Source* (data): BP (2012).

ette payments for all the road traffic (particularly for freight transport), but the main source of funding originates from the taxes placed on used fuel. For example, in our research environment, Finland, fuel taxes (and especially diesel) increased in the year turn of 2011–2012. This was negatively received with road freight transport sector, where diesel prices increased nearly 10%. Change was not the only one in recent years, and taxation increases only result on even higher diesel costs for oil dependent transportation sector. However, during the price hike process competitiveness has increased for more environmentally friendly, alternative power (electricity and gas) using, and energy efficient options.

## 3. Research methodology

This study concentrates on dry ports in a Finnish context. In Finland, competition in the railway sector has been liberated, but currently there is only one company (Finnish Railways), which provides actual transportation services. Two other companies are about to start their operations rather soon (have official permissions more or less granted; Proxion Train and Ratarahti), but of course suitable rolling stock and trustworthy customer base side is still somewhat unclear. As such, Finland can be regarded to be operated by a monopoly company in rails. Still, it is possible that the price of railway transportation is more highly correlated with road transportation compared to a situation with at least modest intra-modal competition. Other important factor in Finland is the well electrified main railway lines. The entire heavily operated railway lines in Finland are electrified and as such are operated by electric trains. These tend to have both lower emissions and lower operating costs than diesel trains. As such, in this paper we concentrate only on electric trains.

### *3.1. Discrete-event simulation model*

In this research we utilize discrete-event simulation (DES). According to Robinson (2004, p. 4) simulation is "Experimentation with a simplified imitation (on a computer) of an operations system as it progresses through time, for the purpose of better understanding and/or improving that system". In DES individual events occur one at a time. Usually DES utilizes servers and queues to analyze behavior of systems (Banks et al., 2005). Simulation can be said to be a constructive approach, where researchers produce solutions to specific problems (Kasanen et al., 1993). In this case, we want to analyze two different transportation systems. It would not be possible to conduct experimentation with the real system (the whole Finnish transportation sector) and as such, either an analytical model or simulation is required. As different parts of railway systems are owned by different entities in many countries (Thompson, 2003), a functional, intermodal transportation system will require co-operation from many different stakeholders. As it is vital to achieve buy-in from all of these different stakeholders, simulation is a better choice than analytical models. According to Robinson (2004), a visual simulation is a good way to com-

municate ideas to others and makes buy-in easier to achieve. Banks et al. (2005) also point out that simulations should be used, when the purpose is to experiment new designs or policies. The possibility to explore all of the different options with a visual representation makes simulation more useful for decision-making than other analytical models.

### 3.2. Data collection and description

The simulation model is based on earlier work of Henttu and Hilmola (2011). The model consists of 193 municipalities in Finland. The total amount of population in these municipalities is over 83% of total population in Finland. During the simulation each new transportation need represents a container, which is going to a specific city. The actual city is randomized and depends on the amount of population in the city, e.g. if 10% of the model's population is in city A, there is a 10% change that the cargo goes to that city or leaves from that city.

When the city of the cargo is known, the model calculates whether a dry port is used or should the cargo go directly to the sea port. Eqs. (1)–(4) present the mathematical decision made in the model, while Table 2 provides the description of the variables. The user can choose whether to minimize emissions (1) or costs (2). In the model both the emissions and costs depend almost totally on ton-kilometers (3 and 4). There are additional emissions in a direct sea port connection due to higher amount of waiting at the sea port compared to a dry port (1). On the other hand, there is additional handling cost in the dry port as the cargo needs to be shifted from road to rail (2). When each individual transportation unit makes their decision, they look through all the possible sea ports and dry ports. Thus, the cargo might use a different sea port depending it will take a direct connection or use a dry port. Eqs. (1)–(4) represented below are created by the authors of this article. Variables chosen for the computational model are based on different sources (e.g. Finnish Transport and Logistics, 2013; Finnish Transport Agency, 2013; Sahin et al., 2009; Quinet and Vickerman, 2004).

$$TE = \sum_{m=0}^{m=n} E_{m,l,d} + \sum_{m=0}^{m=n} I_{m,d} \tag{1}$$

$$TC = \sum_{m=0}^{m=n} C_{m,l,d} + \sum_{m=0}^{m=n-1} CS_{m_1,m_2} \tag{2}$$

$$E_{m,l,d} = W_t \times D_{m,l,d} \times ME_m \tag{3}$$

$$C_{m,l,d} = W_t \times D_{m,l,d} \times MC_m \tag{4}$$

As it can be noticed from equations one through four, there are many different issues, which impact whether a dry port should be used or not. In this paper we will analyze two situations; in first companies minimize their costs, and in second, emissions are minimized. Example of the cost components for Eq. (2) is presented in Fig. 3. There are three different routes in the figure, denoted with numbers 1, 2, and 3. Route 1 is a direct truck connection from Jyväskylä (located in the upper left corner of the figure) to Kotka (located at the bottom of the figure). In this case, the total route cost TC would be $C_{\text{truck, Jyväskylä, Kotka}}$. The route 2 and 3 are part of the total route from Jyväskylä to Kotka by using a dry port in Mikkeli (located in the right side of the figure). The route 2 utilizes a truck from Jyväskylä to Mikkeli, while route 3 is using a train from Mikkeli to Kotka. The total route cost TC for this route is $C_{\text{truck, Jyväskylä, Mikkeli}} + C_{\text{train, Mikkeli, Kotka}} + \text{CS}_{\text{truck, train}}$. Many issues impact both the cost and emission calculations and these parameters are presented in Table 3.

The used truck is a semi-trailer truck with container platform (LIPASTO, 2009). Average weight of loaded container and semi-trailer combination are gathered from LIPASTO (2009), since emission amounts originating from container train and semi-trailer truck are based on the same source. The marginal cost of road and railway transport depends on many issues, but in this research we concentrate on the price of diesel and electricity. Shifting costs between rail and road are based on a case study among two Finnish logistics companies. Additional information about the cost calculations can be found in Henttu

**Table 2**
Variables used in the model.

| | |
|---|---|
| TC | Total costs of transport |
| TE | Total emission of transport |
| $C_{m,l,d}$ | Cost of using modal choice $m$ from location $l$ to destination $d$ |
| $\text{CS}_{m1,m2}$ | Cost of shifting container from transport $m_1$ to $m_2$ |
| $W_t$ | Weight per container in transportation unit $t$ |
| $D_{m,l,d}$ | Distance using modal choice m between location $l$ and destination $d$ |
| $\text{MC}_\text{m}$ | Marginal cost of using modal choice |
| $E_{m,l,d}$ | Emissions using modal choice $m$ from location $l$ to location $d$ |
| $I_{m,d}$ | Emissions due to waiting using modal choice $m$ in destination $d$ |
| $\text{ME}_m$ | Marginal emissions of using modal choice $m$ |

and Multaharju (2011) and Henttu et al. (2010). Both the cost (3) and the environmental calculations (4) depend heavily on ton-kilometers. With truck transportation these can easily be calculated by multiplying the amount of kilometers with the weight of the truck and cargo. In the case of train transportation, the weight is multiplied with the share of the weight of the whole train. With a 40 foot container there are 30 containers in one train and as such, the weight of the locomotive is divided with 30. Eq. (5) shows an example of the calculations of $CO_2$ emissions without using a dry port between Jyväskylä and Kotka while Eq. (6) shows the same calculations by using Mikkeli as a dry port.

$$\begin{aligned} TE &= E_{\text{truck,Jyväskylä,Kotka}} + I_{\text{train,Kotka seaport}} = D_{\text{truck,Jyväskylä,Kotka}} \times W_{\text{truck}} \times ME_{\text{truck}} + I_{\text{train,Kotka seaport}} \\ &= 245\ \text{km} \times (15\ \text{tons} + 15\ \text{tons}) \times 0.044 \frac{\text{kg}}{\text{tkm}} + 3\ \text{kg} = 323.4\ \text{kg} + 3\ \text{kg} = 326.4\ \text{kg} \end{aligned} \tag{5}$$

$$\begin{aligned} TE &= E_{\text{truck,Jyväskylä,Mikkeli}} + E_{\text{train,Mikkeli,Kotka}} + I_{\text{truck,Jyväskylä,Mikkeli dryport}} \\ &= D_{\text{truck,Jyväskylä,Mikkeli}} \times W_{\text{truck}} \times ME_{\text{truck}} + D_{\text{train,Mikkeli,Kotka}} \times W_{\text{train}} \times ME_{\text{train}} + I_{\text{truck,Mikkeli dryport}} \\ &= 113\ \text{km} \times (15\ \text{tons} + 15\ \text{tons}) \times 0.044 \frac{\text{kg}}{\text{tkm}} + 166\ \text{km} \times \left(15\ \text{tons} + 30 \frac{\text{tons}}{\text{platform}} + \frac{83 \frac{\text{tons}}{\text{locomotive}}}{30 \frac{\text{containers}}{\text{locomotive}}}\right) \\ &\quad \times 0.0067\ \frac{\text{kg}}{\text{TKm}} + 1.5\ \text{kg} \\ &= 149.16\ \text{kg} + 53.13\ \text{kg} + 1.5\ \text{kg} = 203.8\ \text{kg} \end{aligned} \tag{6}$$

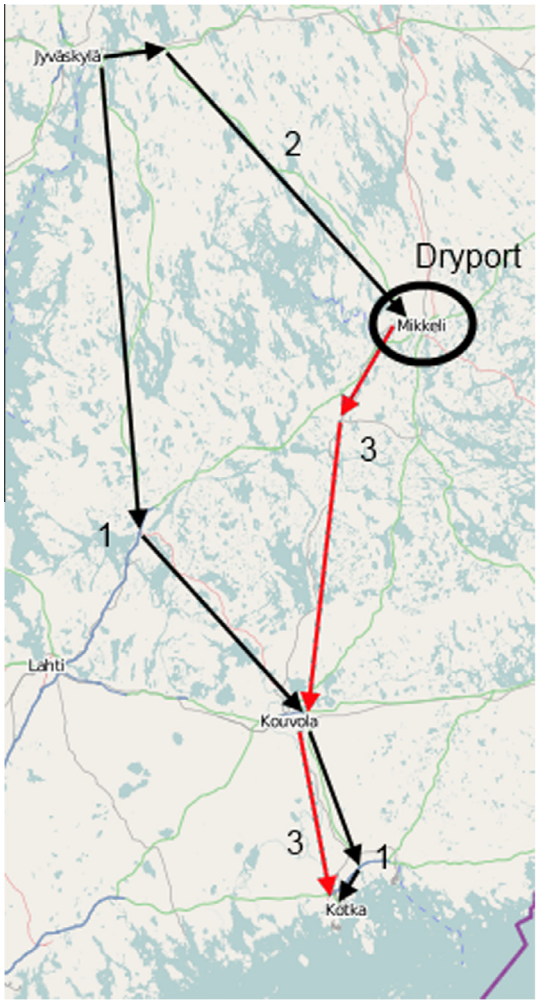



**Fig. 3.** Example of route cost components for two different routes from Jyväskylä to Kotka. Map from OpenStreeMap, © OpenStreetMap contributors, CC-BY-SA license, www.openstreetmap.com.

**Table 3**
Parameters used in the model.

| | |
|---|---|
| Average weight of container | 15 tons |
| Container | 40 foot |
| Average weight of truck and platform | 15 tons |
| Weight of locomotive | 83 tons |
| Weight of train platform | 30 tons |
| Platforms per train | 30 platforms per locomotive |
| Shifting cost between truck and train | 35 € |
| Marginal cost of road transport | 0.04604–0.0527 €/tkm |
| Marginal cost of railway transport | 0.022–0.02396 €/tkm |
| Emissions due to waiting in sea port (truck) | 15 min; 1.5 kg $CO_2$ |
| Emissions due to waiting in dry port (truck) | 30 min; 3 kg $CO_2$ |
| Emissions of container train | 6.7 g $CO_2$/tkm |
| Emissions of semi-trailer truck | 44 g $CO_2$/tkm |

The DES model utilizes network-based modelling. In network-based modelling a network consisting of vertices and edges connect different parts of the model together. Distance between the locations and the average moving speed can then be used as delays in the actual DES structure. There are four different "groups" in the DES system: direct imports, direct exports, dry port imports, and dry port exports. The amount of imports and exports are evenly distributed, but the share of dry port usage depends on Eqs. (1) and (2). The simulation model calculates the share of locations willing to use a dry port and then balances the arrival table for the DES model. The four groups are presented in Fig. 4.

The simulation model combines the DES model with a graphical representation of the Finnish road and railway network. The trucks and trains follow the actual roads and railway tracks, which improves visualization over the problem domain. Example of the model during run-time is presented in Fig. 5.

The graphical user interface allows the user to make changes into the model. Most of the decisions will impact the actual structure of the network during the simulation run. As the simulation model also has the representation of the whole network embedded in a map, the user can more easily notice, what changes occur in different parts of the network, if changes are made to the decision parameters. The options of the simulation model are presented in Table 4.

The warm up period for the simulation is a relatively short one (matter of hours) as the biggest additional delays (in addition to the actual driving times) are generated when the trucks loads are batched to become a whole train load. As a full train

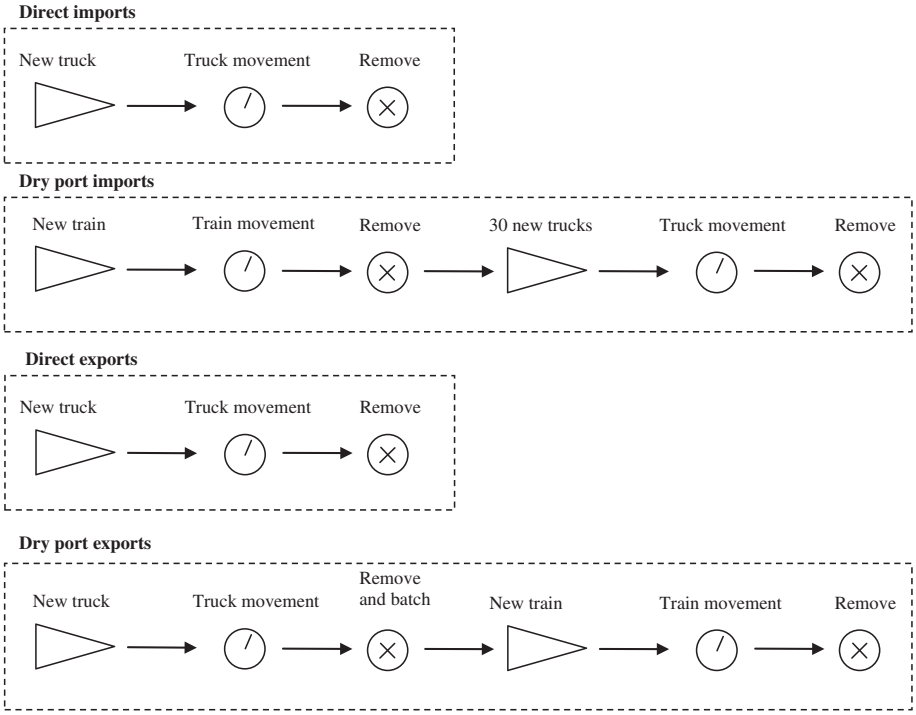



**Fig. 4.** The DES models for different types of transports.

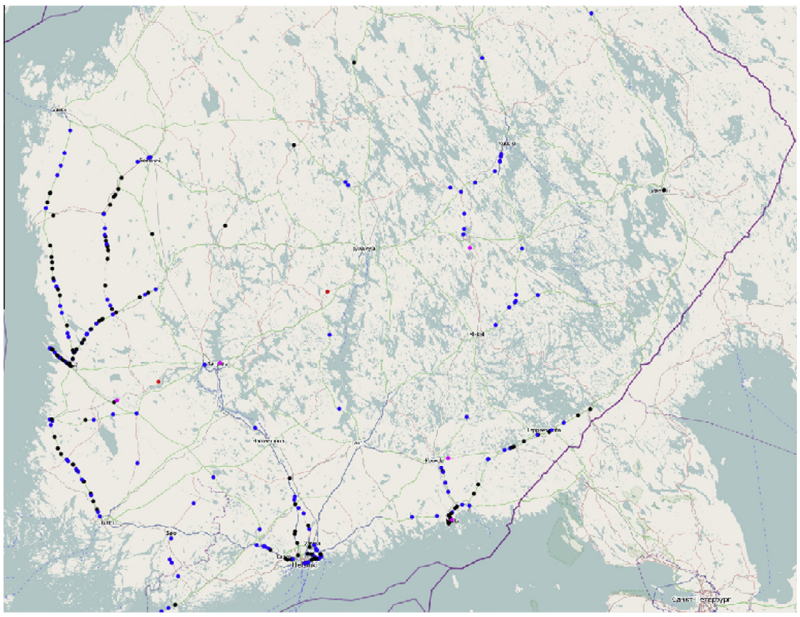


**Fig. 5.** Example of the graphical overview of the simulation model during run time. Map from OpenStreeMap, © OpenStreetMap contributors, CC-BY-SA license, www.openstreetmap.com.

**Table 4**
Simulation model options.

| |
|---|
| Minimize costs or minimize emissions |
| Cost components |
| Locations of dry ports |
| Size of container |
| Weight of container |
| Direct connections allowed/disallowed |

load requires 30 containers (40″) and the model generates 120 trucks per hour which use up to 20 different dry ports (6 trucks for each dry port), the first full train loads require on average about 5 h (30 containers divided with 6 containers per hour). The whole simulation run used in the analysis was 1000 h and as such, the potential warp up period is almost miniscule.

## 4. Results of the simulation model

Model compares two different transportation scenarios in Finland. First one uses only road transportation, while in the latter one it is possible to use dry ports with hinterland rail transport. The simulation model minimizes either total transportation costs or total emissions in the latter system. Aim of the model is to find out, if there are benefits of dry ports in terms of transportation costs or $CO_2$ emission amounts in Finnish transportation network. Price levels of electricity and diesel fuel start from levels, which are below current actual price levels in Finland (0.01 €/kW h and 0.8 €/l respectively). Road transport vehicles use diesel fuel as source of energy, and the price of diesel for heavy vehicles is almost the same as for private consumers. A small discount is possible for road transport companies due to procurement volume of diesel fuel and possibility to use summer diesel also during winter season i.e. the price of diesel fuel is about 1.4–1.6 €/l, if VAT is included. Road transport companies can deduct VAT from the diesel price. VAT excluded price for road transport companies is currently about 1.13–1.30 €/l. Price levels increase up to 100% of the starting values (up to 0.02 €/kW h and 1.6 €/l). Used trains between sea ports and dry ports are electrified.

Fig. 6 shows the optimal amount of cargo utilizing dry ports with different electricity and diesel fuel price levels (diesel fuel prices without VAT). Price of electricity affects only the scenario, which allows the usage of dry ports, since trains travelling between dry ports and sea ports are assumed to be electric trains. Price of diesel affects both scenarios, since road transport is used in both.

If electricity is expensive (0.2 €/kW h) and price of diesel is at model's minimum level (0.8 €/l), then usage of dry ports is minimized in the model (∼30%). It means that approximately 30% of cargo is transported through dry ports and all the other

volumes are transported with unimodal road transport directly between cities and sea ports (this level is not even near of the current situation of Finnish sea ports, where at best 5–10% from container volumes are transported by rail). If price of electricity is minimized and price of diesel is maximized in the model, then the usage of dry ports is maximized and about 41% of freight is transported through dry port structure. According to the simulation model, 30–41% of cargo should be transported using dry ports in Finland to minimize transportation costs. Larger drops in Fig. 6 means that some large city (large cities have large cargo volumes) finds out that it is cheaper to use unimodal road transport due to changed price levels and decides to stop using dry port. This can happen, when price of electricity is expensive enough and price of diesel is cheap enough. If the price of diesel is 1.2 €/l or higher, then the usage of dry ports is almost 40% or even higher, if the price of electricity reaches its maximum possible value.

As there are no differences in the cost between imports and exports, there are no differences between the share of direct connections between imports and exports. Every location will use a direct connection for both the imports and exports. Due to this reason the share of imports and exports also does not have an impact on the results of the model. It is possible that by including information about the weight of the cargo, as well as delivery due dates, the imports and exports would differ in their profiles on the usage of dry ports.

Fig. 7 clarifies difference in $CO_2$ emission amounts between dry port implemented transportation network and unimodal road transportation scenario. Variables are again price of electricity and price of diesel.

Results of Fig. 7 are shown as emission difference between dry port implemented and unimodal road transport scenarios as kilograms per one container. The model calculates average emission amount for one container for both scenarios. The less dry ports are used, the more road transport is utilized, and the more $CO_2$ emissions are created in the dry port scenario. Though, dry port implemented scenario is environmentally friendlier with all different price levels for electricity and diesel. Shape of Fig. 7 is similar to shape of Fig. 6. Reason for this is that large drops in cargo using dry ports in Fig. 6 are due to some large volume cities decide to stop using dry ports, and starts using only unimodal road transport, which emits more $CO_2$ emissions than rail transport. These drops can be seen in Fig. 7 as decreased difference of $CO_2$ emissions between two reviewed scenarios. Every time the model chooses to drop one city out, the amount of emissions increase. Table 5 provides a summary of the results with various railway and road costs.

Minimum difference of $CO_2$ emission between two researched scenarios is about 47 kg per one container. This difference occurs with the most expensive electricity and cheapest configuration for diesel. These settings lead to a situation, where the least number of cities use dry ports. If only sea ports are used, 145 kg of $CO_2$ is emitted per container and as such, the dry port system provides impressive decreases in emissions. It should be noted that there exists small difference between the scenarios, where road transportation costs are ranging from medium (1.2 €/l) to high (1.6 €/l). On the other hand, the cost of railway transportation has a big difference in the scenario, where the price of road transportation is small (0.8 €/l). Due to the decreasing supply of oil it is unlikely that the price of diesel will significantly decrease in the future. Table 6 shows the cost differences of transporting a container using a dry port system instead of driving directly to sea ports.

The cost differences vary between 7.96 € to 24.80 € per container. It should be noted, that as the truckers are minimizing their costs, the dry port scenario will always have lower costs as they can also choose a direct connection to sea port, if it is

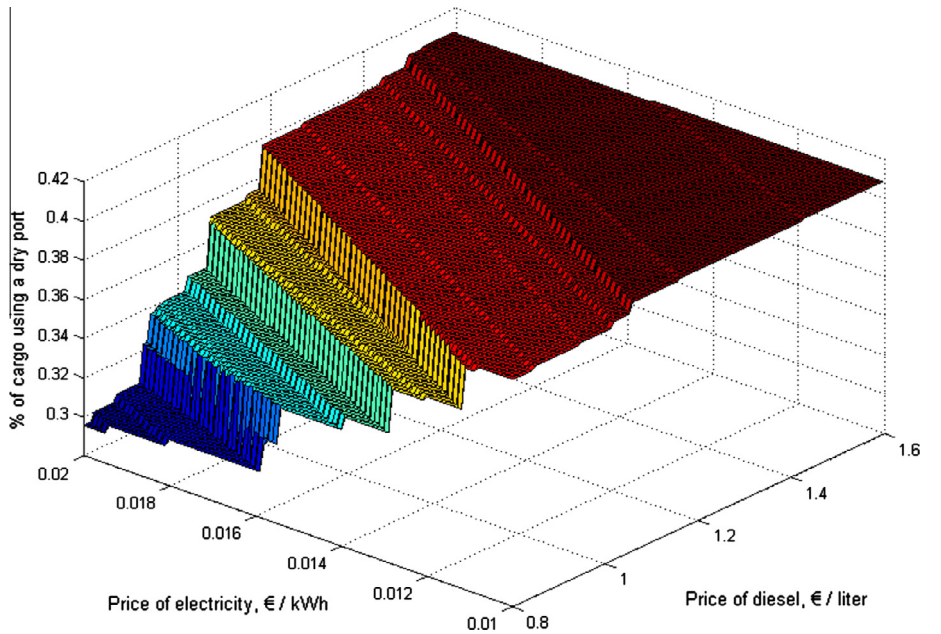



**Fig. 6.** Percentage of cargo using dry ports.

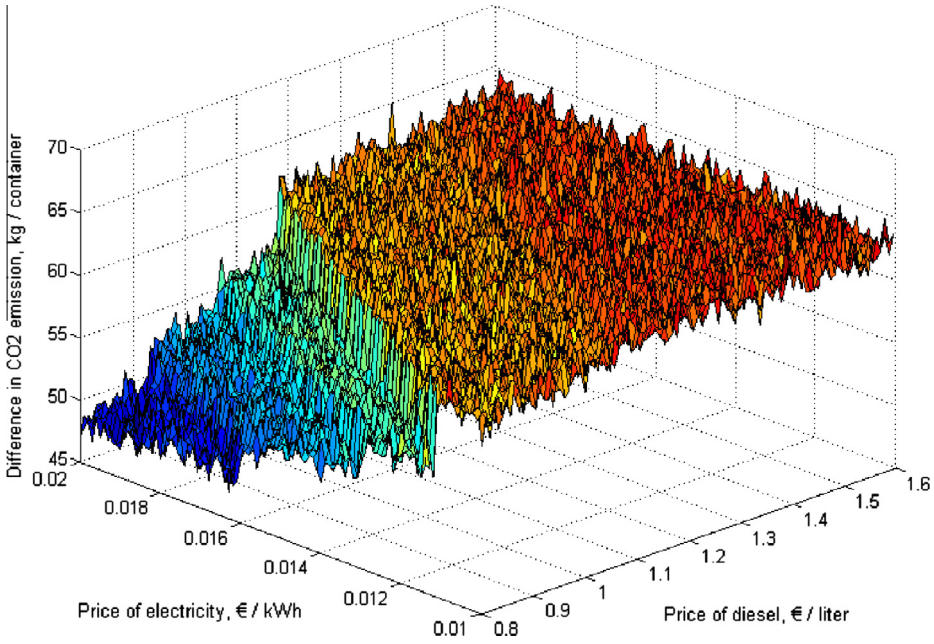



**Fig. 7.** Difference of $CO_2$ emissions between dry port implemented transportation and unimodal road transportation scenario.

**Table 5**
Summary on the difference of $CO_2$ emissions (kg) with different transportation costs between the intermodal and unimodal transportation systems.

| | Price of diesel | | | | |
|---|---|---|---|---|---|
| | 0.8 €/l | 1.0 €/l | 1.2 €/l | 1.4 €/l | 1.6 €/l |
| *Price of electricity* | | | | | |
| 0.01 €/kW h | 61.13 kg | 59.81 kg | 62.43 kg | 62.21 kg | 63.81 kg |
| 0.0125 €/kW h | 57.55 kg | 63.21 kg | 62.95 kg | 63.34 kg | 63.21 kg |
| 0.015 €/kW h | 52.57 kg | 61.88 kg | 59.58 kg | 62.66 kg | 63.46 kg |
| 0.0175 €/kW h | 50.27 kg | 57.43 kg | 62.46 kg | 63.12 kg | 62.06 kg |
| 0.02 €/kW h | 47.00 kg | 51.81 kg | 60.36 kg | 65.09 kg | 61.98 kg |

**Table 6**
Summary on the difference of transportation costs (€/container) with different transportation costs between the intermodal and unimodal transportation systems.

| | Price of diesel 0.8 €/l | 1.0 €/l | 1.2 €/l | 1.4 €/l | 1.6 €/l |
|---|---|---|---|---|---|
| *Price of electricity* | | | | | |
| 0.01 €/kW h | 13.20 € | 15.33 € | 18.89 € | 21.91 € | 24.80 € |
| 0.0125 €/kW h | 12.10€ | 15.23 € | 18.19 € | 20.89 € | 23.55 € |
| 0.015 €/kW h | 10.40 € | 13.70 € | 15.85 € | 19.63 € | 21.92 € |
| 0.0175 €/kW h | 9.66 € | 12.23 € | 15.66 € | 18.14 € | 20.51 € |
| 0.02 €/kW h | 7.96 € | 10.24 € | 14.02 € | 17.66 € | 19.54 € |

cheaper. Also, as the costs are average costs, in some cases the benefits are small ones for the truckers, but some other operators could benefit greatly. The cost differences are also linear compared to the differences in emissions, which have big steps in the differences. Fig. 8 shows the locations of the dry ports, their expected maximum demand, variation in their demand, as well as the demand for railway transports. More exact values for individual dry ports are presented in Table 7.

As it is possible to notice from Fig. 8 and Table 7, many of the dry ports have a stable demand between the scenarios (a small amount of variation). Tampere and Kouvola could work as good dry ports, which could consolidate cargo flows from other inland intermodal terminals as well. Tampere and Jyväskylä could handle over 10% of all of the cargo in the model and they have a relatively small amount of variation. Kouvola, on the other hand, has a relatively high amount of variation in

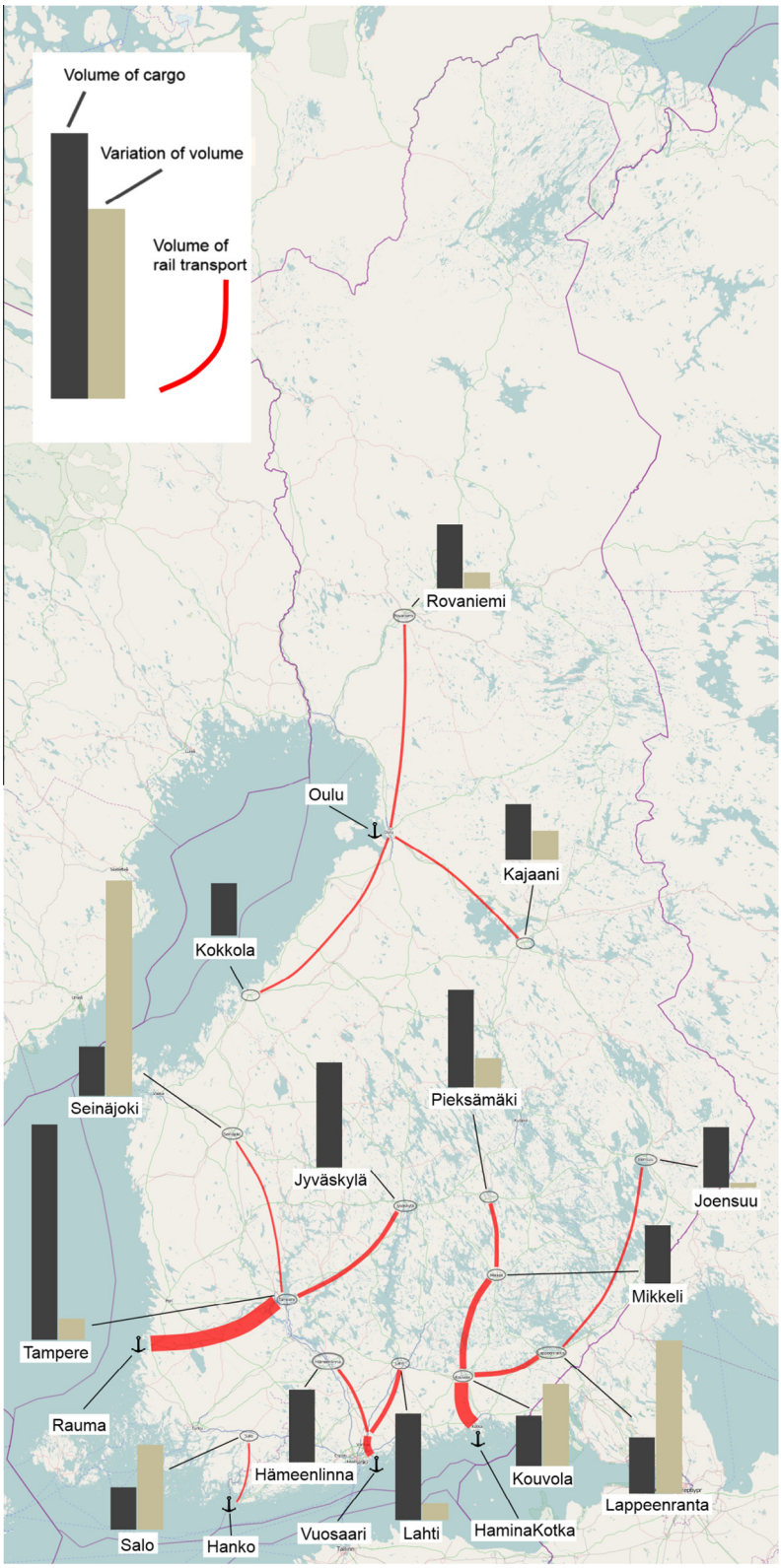



**Fig. 8.** Locations for potential dry ports and their potential volumes. Map modified from OpenStreeMap, © OpenStreetMap contributors, CC-BY-SA license, www.openstreetmap.com.

**Table 7**
Volume of potential dry ports, demand variation between scenarios and used sea port.

| City | Share of cargo (%) | Coefficient of variation (%) | Sea port |
|---|---|---|---|
| Tampere | 7.75 | 5.13 | Rauma |
| Lahti | 3.85 | 4.05 | Vuosaari |
| Jyväskylä | 3.78 | 0.00 | Rauma |
| Pieksämäki | 3.50 | 7.07 | HaminaKotka |
| Hameenlinnä | 2.61 | 0.00 | Vuosaari |
| Rovaniemi | 2.29 | 3.83 | Oulu |
| Joensuu | 2.18 | 1.17 | HaminaKotka |
| Mikkeli | 2.10 | 0.07 | HaminaKotka |
| Kajaani | 2.02 | 7.12 | Oulu |
| Lappeenranta | 2.01 | 37.29 | HaminaKotka |
| Kokkola | 1.89 | 0.00 | Oulu |
| Kouvola | 1.83 | 19.93 | HaminaKotka |
| Seinäjoki | 1.75 | 52.33 | Rauma |
| Salo | 1.53 | 20.67 | Hanko |

**Table 8**
Summary on the difference of transportation costs (€/container) with different transportation costs between the intermodal and unimodal transportation systems when emissions are minimized.

| | Price of diesel | | | | |
|---|---|---|---|---|---|
| | 0.8 €/l | 1.0 €/l | 1.2 €/l | 1.4 €/l | 1.6 €/l |
| *Price of electricity* | | | | | |
| 0.01 €/kW h | −5.47 € | −1.79 € | 2.23 € | 6.37 € | 9.36 € |
| 0.0125 €/kW h | −6.24 € | −3.40 € | 1.42 € | 5.08 € | 9.16 € |
| 0.015 €/kW h | −9.03 € | −4.88 € | −1.11€ | 1.95 € | 5.36 € |
| 0.0175 €/kW h | −11.11 € | −6.45 € | −2.79 € | 0.06 € | 4.36 € |
| 0.02 €/kW h | −12.57 € | −8.81 € | −5.35 € | −0.71 € | 3.99 € |

demand between scenarios. Kouvola could initially become a dry port to consolidate flows from Pieksämäki, Mikkeli, and Joensuu. Even if the local operators in the Kouvola region could not benefit financially from the dry port, it would be easy so start accepting local goods in the future, if the cost structures change.

In Southern Finland Hämeenlinna and Lahti could as well function as inland terminals. The terminals connect near Kerava, which is already a major logistical center in Finland. The volumes from Kerava (6.46%) are smaller than from Tampere (13.28%) or Kouvola (11.62%), but the logistical infrastructure is already very good. The final dry port in Southern Finland, Salo, has a relatively high variation and a small amount of cargo. As such, it might not be a good location for a dry port.

In Northern Finland the sea port of Oulu could be served from three dry ports: Rovaniemi, Kajaani, and Kokkola. The individual dry ports would have relatively small volumes so the question remains whether enough cargo could be generated. It might be that serving one individual dry port in Northern Finland is not cost efficient for the sea port, but serving all three of them would be.

The previous analyses concentrated on analyzing the situation where individual actors minimize their own costs. If $CO_2$ emissions are minimized instead of transportation costs, then the difference in price of one container between two researched scenarios evolves differently (Table 8).

If the price of electricity is maximum and price of diesel is minimum, then average transportation costs per one container are 12.57 € more expensive than the average costs in unimodal road transport scenario i.e. dry port implemented scenario is more expensive. Price difference changes linearly, if price of electricity decreases and price of diesel fuel increases. The dry port system is able to have lower costs in many cases, but this emission minimization scenario is mostly unrealistic as organizations tend to minimize their costs and can be used to understand what would be the lowest possible emissions achievable using a dry port system to the fullest.

## 5. Sensitivity analysis on estimate emission payments per container

Challenge with $CO_2$ emissions in transportation is that their reduction in radical manner is not possible at transportation mode level (e.g. trucks or short sea shipping), but it is merely question of avoiding using mineral fuels and replacing them with electricity. Latter one ought to be produced with environmentally (will not produce $CO_2$ or very low amounts) sustainable manner, like originating e.g. from solar power, wind mills, hydropower or nuclear power plants in order to have maximum utility. Railway also beats road option in a manner that it requires lower amount of land area to transport same

**Table 9**
Estimated emission payments per container with varying $CO_2$ price and emission difference.

| | $CO_2$ emission difference (kg/container) | | | |
|---|---|---|---|---|
| | 50 | 55 | 60 | 65 |
| *$CO_2$ price (€)* | | | | |
| €4.00 | €0.20 | €0.22 | €0.24 | €0.26 |
| €7.60 | €0.38 | €0.42 | €0.46 | €0.49 |
| €20.00 | €1.00 | €1.10 | €1.20 | €1.30 |
| €60.00 | €3.00 | €3.30 | €3.60 | €3.90 |
| €100.00 | €5.00 | €5.50 | €6.00 | €6.50 |
| €120.00 | €6.00 | €6.60 | €7.20 | €7.80 |
| €180.00 | €9.00 | €9.90 | €10.80 | €11.70 |

amount of freight transported – giving more space for forests and plants to act as world's carbon sink (Henttu et al., 2010). This positive effect was not taken into account in our simulation experiment.

Even if long-term forecasts for emission right costs of $CO_2$ are really high, in the emission exchange ICE-ECX (2012) their price has continued to decline. Actually it is so that $CO_2$ emission contract prices are flirting in the lowest levels, since the trade started in year 2006. Current average contract (futures contracts) price is just slightly above 7.6 Euros per ton of $CO_2$ emitted (in November 2012). Of course major factor for such low value is the macro economic situation in Europe (recession and in some country cases depression), but also could be assumed that implementation of emission payments on heavy industries, and the dawn of these payments in transportation sector have changed the behavior of for-profit decision makers. It could be assumed that new emission free or significantly removing technologies have been invested, but in some cases heavily polluting factories have been offshored and/or outsourced to outside of EU region. Currently within European Union area emission trading ($CO_2$) is everyday life already in energy production, heavy industries and airlines (although, not all emissions are needed to be purchased from exchange as some fixed amounts are given for free as emphasis is on emission control and steady reduction of it).

Based on DG Regional (2008) and Maibach et al. (2008) highest trajectory of $CO_2$ costs in European Union should be roughly 10 times of that, what they are sold at exchange today. Also their growth ought to continue until 2050, when they should reach 180 Euros per $CO_2$ emitted ton. High level of costs is justified with discontinuities in climate change processes – as emerging economies are not that interested from $CO_2$ prevention, additional emissions produced in developed economies have considerably larger effects. So, marginal costs are then relatively high and do not follow incremental cost development. However, despite all of bleak and high cost forecasts of the future, fact is that today $CO_2$ emissions are trading at lowest point possible in all scenarios proposed by DG Regional (2008) and Maibach et al. (2008). Actually these publications forecasted that around seven Euros was to be the lowest possible value in year 2010. Recently we have received opinions from experts of emission trading that EU should radically decrease the amount of sold emission rights in order to secure greener future. With current levels e.g. a bit greener coal in energy production is better alternative than other less polluting options, like gas (Evans, 2012). However, it should be reminded that there exists research works, which argue that in Europe $CO_2$ ton cost should be around 4–20 Euros (Quinet and Vickerman, 2004) in the early period of $CO_2$ reduction adaption (pre 2020 world). So, maybe current rather lowly valued emission cost is just reflecting easier demands before the year 2020, but this will drastically change as we proceed towards ambitious goal of year 2050.

In our simulation experiment it was assumed that $CO_2$ emissions do not cost at all – so in other words we have not implemented trading scheme. However, within EU area this is already practice in aviation, and trade partners such as USA, China and Russia have clearly stated to object the new system, where all planes landing to EU area are in the sphere of this system. We could assume that this trading scheme will enlarge to other transportation sub-sectors, like road, sea and rail. Therefore, using mineral fuels becomes double edged sword; higher prices of diesel oil are not the only matter to be concerned, but also higher priced $CO_2$ emissions should be taken into account. It should be noted that diesel consumption and $CO_2$ emissions go hand in hand, so decision makers in emission trading scheme will really have a challenge to deal with. In Table 9 we have estimated cost effects of this possible change with varying emission differences between road and dry port network. As could be noted, cost significance to our research work analyses appears around 20–60 € per ton area. Thereafter, nearly every scenario simulated will be at least having positive result out of using $CO_2$ minimization strategy whatever the electricity and diesel price is (Table 8), and financial performance of best scenarios approaches cost minimization strategy (Table 6).

## 6. Conclusions

In this paper the first main research question was: "How can the locations of potential dry ports be estimated on a regional scale?" As Rodrigue et al. (2010) point out, a dry port needs to have enough economies of scale to justify its existence. At the same time, shippers need to be able to operate with lower cost-structure in order to utilize a dry port system (Cullinane and Wilmsmeier, 2011). In this research we provided a methodology to construct a DES simulation model, where shippers minimize their own costs and a potential dry port system emerges from the results of the model. The model is able to

give an upper bound for the potential traffic flows, which allows the decision-makers to concentrate their efforts on the most promising regions. The results of the model can be used at a regional, national, or even international level (if the regional area would cover multiple countries) to help devise strategies to create efficient transportation corridors. This will also help minimizing overinvestment to individual dry ports (Bergqvist and Wilmsmeier, 2009; Rodrigue and Notteboom, 2012).

The first sub-research question: "How can the stability of demand for dry ports be estimated?" was analyzed by varying the main cost-elements (truck and railway transport costs). These are the most important cost-elements in a dry port system (Kim and Van Wee, 2011). With the help of the simulation model it is possible to estimate, how much the demand changes in different scenarios. Locations with a high volume of cargo and a small amount of variation are prime candidates for a dry port as they have enough massification even in different types of scenarios (Rodrigue et al., 2010).

The second sub-research question: "How do potential dry ports interact in larger logistical systems?" could also be analyzed with the help of the simulation model. As dry ports are part of larger logistical systems (Roso et al., 2009), it is important to understand the ramifications of individual dry ports and the connectivity of different sea ports to hinterland transportation networks. The inland terminals can also have different kind of roles (Rodrigue and Notteboom, 2012) and it is important to understand how different locations are connected.

The second main research question was: "How much can the emissions of transportation be lowered by utilizing dry ports?" According to developed simulation model, 30–41% of Finnish inland transportation should be transported by using dry ports to minimize transportation costs of the whole Finnish inland transportation. From other angle, 59–70% of the inland transportation should be transported by unimodal road transportation, because using dry ports would increase costs. These results are similar to the findings of Roso (2007), where a dry port was able to decrease $CO_2$ emissions by 25%. Decision-makers should also understand, that while costs might increase linearly, the potential environmental benefits of dry ports increase in steps. A small change in transportation costs can have a big impact on emissions as a cluster might start using a dry port system, when the break-even distance is broken. The break-even distance is a complex issue (Kim and Van Wee, 2011) and needs to be carefully studied by the decision-makers with advanced tools.

According to the results of this paper, by implementing dry port network in Finnish transportation context it will decrease $CO_2$ emissions, if compared to transportation network, which utilizes only road transport. Even though $CO_2$ emissions can be decreased by up to 45% with implemented price settings, the actual benefit in monetary terms is under one percent of total transportation costs, if one ton of $CO_2$ is assumed to cost 20 € (currently price in exchange is 7.6 € per ton). Furthermore, total costs of transportation can be decreased by using dry port network. Average cost of transporting one container in unimodal road transport scenario is ca. 160 €. Depending on the price level of electricity and diesel fuel, transportation costs of one container in dry port implemented scenario costs 135–153 € i.e. percentual cost savings are about 4–16%.

Average amount of $CO_2$ emissions per container in scenario, in which only road transport is utilized, is 145 kg i.e. amount of $CO_2$ emissions per container can be decreased using dry ports by 32% to 45% depending on the price levels of electricity and diesel fuel. Although $CO_2$ decrease is proportionally considerable, the monetized benefit is almost nonexistent with current $CO_2$ price level (up to 20 € per ton). Situation will change, if price level of $CO_2$ increases radically. This was illustrated within discussion section. Basically companies in high $CO_2$ price environment could just minimize $CO_2$ emissions and economy would follow. Explanation is coming from the connection of $CO_2$ emissions and diesel oil used – together they will take largest part from total costs as $CO_2$ emission costs are increasing from 20 € per ton to 60 € and especially beyond.

Currently the model has some limitations. One of the limitations is that it is based on population as there is currently no information regarding transportation flows within Finland. Secondly, there are currently no capacity limitations anywhere in the model. This tends to be more of an issue for railways, but it should be mentioned, that the railway lines used by the simulation model are currently not in full utilization. Thirdly, all of the cargo is assumed to be similar. Even if one only concentrates on containers, there will be differences in the weight of containers. Also, some containers might require a faster connection to a sea port in which case dry ports are not an option. Due to these reasons the model can only give an upper bound for the potential cargo flows. Furthermore, waiting times in dry port terminals are not taken into account, since comparison of delivery times was not aim of this research work. Adding data regarding both the weight and the delivery times could also have an impact on the differences between usage of dry ports between imports and exports, e.g. a dry port could provide more benefits for either imports or exports.

Further research includes expansion of the model to include more exact information from the cargo flows. One possibility is to use industrial production in various cities instead of amount of population. Secondly, capacity should be included in the model. This could be a maximum capacity for a dry port (containers/year) or a maximum capacity of trucks for a sea port. This is the case in some European sea ports (such as Rotterdam). Another interesting avenue for further research would be the analysis of "optimal" price for $CO_2$ emissions. In the model the shippers could bid on the amount of emissions they are going to emit with their current transportation solution. As such, the model will reach an equilibrium price for emissions which can be gapped to a pre-defined level. This kind of analysis could be used by decision-makers to estimate how much shippers are willing to pay for emission rights. In order to reach a low carbon economy, the whole society needs to look for innovative solutions. Dry ports are one piece of the solution to reach a safer and cleaner tomorrow.

**Appendix A. European Brent prices and Federal Reserve Bank Credit (18 December 2002–1 August 2012)**

| Date | European Brent (Dollars per Barrel, 200d) | Reserve Bank Credit (USA), in mill. USD |
|---|---|---|
| December 18, 2002 | 25.96 | 692724.00 |
| January 08, 2003 | 26.36 | 697396.00 |
| February 05, 2003 | 26.95 | 694400.00 |
| March 05, 2003 | 27.65 | 702157.00 |
| April 02, 2003 | 28.25 | 708047.00 |
| May 07, 2003 | 28.13 | 708010.00 |
| June 04, 2003 | 28.16 | 716907.00 |
| July 02, 2003 | 28.14 | 719564.00 |
| July 30, 2003 | 28.11 | 713436.00 |
| August 06, 2003 | 28.19 | 715952.00 |
| September 03, 2003 | 28.72 | 725288.00 |
| October 01, 2003 | 28.72 | 727177.00 |
| November 05, 2003 | 28.51 | 725133.00 |
| December 03, 2003 | 28.15 | 734989.00 |
| January 07, 2004 | 28.06 | 743275.00 |
| February 04, 2004 | 28.62 | 732294.00 |
| March 03, 2004 | 29.23 | 736638.00 |
| April 07, 2004 | 29.97 | 733595.00 |
| May 05, 2004 | 30.57 | 742210.00 |
| June 02, 2004 | 31.41 | 749700.00 |
| July 07, 2004 | 32.33 | 755431.00 |
| August 04, 2004 | 33.29 | 758080.00 |
| September 01, 2004 | 34.70 | 764053.00 |
| October 06, 2004 | 36.50 | 764782.00 |
| November 03, 2004 | 38.34 | 773228.00 |
| December 01, 2004 | 39.54 | 785481.00 |
| January 05, 2005 | 40.26 | 793890.00 |
| February 02, 2005 | 41.42 | 786418.00 |
| March 02, 2005 | 42.30 | 784766.00 |
| April 06, 2005 | 44.44 | 783601.00 |
| May 04, 2005 | 45.84 | 786573.00 |
| June 01, 2005 | 46.40 | 792344.00 |
| July 06, 2005 | 48.00 | 795635.00 |
| August 03, 2005 | 48.85 | 799948.00 |
| September 07, 2005 | 51.41 | 799875.00 |
| October 05, 2005 | 53.63 | 800690.00 |
| November 02, 2005 | 55.12 | 802586.00 |
| December 07, 2005 | 56.29 | 814193.00 |
| January 04, 2006 | 56.68 | 833038.00 |
| February 01, 2006 | 57.81 | 814128.00 |
| March 01, 2006 | 58.90 | 820710.00 |
| April 05, 2006 | 60.12 | 820501.00 |
| May 03, 2006 | 61.42 | 823271.00 |
| June 07, 2006 | 62.26 | 829379.00 |
| July 05, 2006 | 62.82 | 836010.00 |
| August 02, 2006 | 64.25 | 832767.00 |
| September 06, 2006 | 66.20 | 832250.00 |
| October 04, 2006 | 66.61 | 829514.00 |
| November 01, 2006 | 66.26 | 833397.00 |
| December 06, 2006 | 66.09 | 842309.00 |
| January 03, 2007 | 66.13 | 859406.00 |
| February 07, 2007 | 64.43 | 841518.00 |
| March 07, 2007 | 63.31 | 851651.00 |
| April 04, 2007 | 62.87 | 852252.00 |
| May 02, 2007 | 62.36 | 866895.00 |

(*continued on next page*)

**Appendix A** (*continued*)

| Date | European Brent (Dollars per Barrel, 200d) | Reserve Bank Credit (USA), in mill. USD |
|---|---|---|
| June 06, 2007 | 61.56 | 858016.00 |
| July 11, 2007 | 62.37 | 853943.00 |
| August 01, 2007 | 63.82 | 857671.00 |
| September 05, 2007 | 65.39 | 856609.00 |
| October 03, 2007 | 66.99 | 861511.00 |
| November 07, 2007 | 70.55 | 864439.00 |
| December 05, 2007 | 73.94 | 866839.00 |
| January 02, 2008 | 76.59 | 891168.00 |
| February 06, 2008 | 79.71 | 861798.00 |
| March 05, 2008 | 82.64 | 873776.00 |
| April 02, 2008 | 85.83 | 875607.00 |
| May 07, 2008 | 90.38 | 869232.00 |
| June 04, 2008 | 95.28 | 879344.00 |
| July 02, 2008 | 101.45 | 890038.00 |
| August 06, 2008 | 107.51 | 891511.00 |
| September 03, 2008 | 109.59 | 895447.00 |
| October 01, 2008 | 110.10 | 1391136.00 |
| November 05, 2008 | 107.23 | 2055824.00 |
| December 03, 2008 | 103.20 | 2117585.00 |
| January 07, 2009 | 96.07 | 2157607.00 |
| February 04, 2009 | 90.09 | 1840589.00 |
| March 04, 2009 | 83.06 | 1891362.00 |
| April 01, 2009 | 74.83 | 2046241.00 |
| May 06, 2009 | 64.63 | 2041311.00 |
| June 03, 2009 | 59.00 | 2064875.00 |
| July 01, 2009 | 55.33 | 1986822.00 |
| August 05, 2009 | 52.82 | 1977894.00 |
| September 02, 2009 | 54.07 | 2063318.00 |
| October 07, 2009 | 56.84 | 2119490.00 |
| November 04, 2009 | 60.27 | 2148875.00 |
| December 02, 2009 | 63.41 | 2187185.00 |
| January 06, 2010 | 66.91 | 2216302.00 |
| February 03, 2010 | 69.25 | 2231328.00 |
| March 03, 2010 | 71.31 | 2262336.00 |
| April 07, 2010 | 73.03 | 2290119.00 |
| May 05, 2010 | 75.13 | 2311761.00 |
| June 02, 2010 | 75.44 | 2319907.00 |
| July 07, 2010 | 75.99 | 2314607.00 |
| August 04, 2010 | 76.90 | 2309095.00 |
| September 01, 2010 | 76.85 | 2286672.00 |
| October 06, 2010 | 77.30 | 2284557.00 |
| November 03, 2010 | 77.86 | 2280713.00 |
| December 01, 2010 | 79.12 | 2317835.00 |
| January 05, 2011 | 80.79 | 2410850.00 |
| February 02, 2011 | 82.18 | 2438487.00 |
| March 02, 2011 | 84.36 | 2518666.00 |
| April 06, 2011 | 89.65 | 2619634.00 |
| May 04, 2011 | 93.83 | 2686873.00 |
| June 01, 2011 | 97.13 | 2770774.00 |
| July 06, 2011 | 101.68 | 2854280.00 |
| August 03, 2011 | 105.32 | 2849519.00 |
| September 07, 2011 | 108.36 | 2840859.00 |
| October 05, 2011 | 110.69 | 2835665.00 |
| November 02, 2011 | 112.33 | 2817608.00 |
| December 07, 2011 | 113.70 | 2797951.00 |

**Appendix A** (*continued*)

| Date | European Brent (Dollars per Barrel, 200d) | Reserve Bank Credit (USA), in mill. USD |
|---|---|---|
| January 04, 2012 | 113.32 | 2900756.00 |
| February 01, 2012 | 112.70 | 2905666.00 |
| March 07, 2012 | 112.83 | 2864857.00 |
| April 04, 2012 | 113.79 | 2843183.00 |
| May 02, 2012 | 114.39 | 2845876.00 |
| June 06, 2012 | 113.65 | 2830902.00 |
| July 11, 2012 | 111.53 | 2849051.00 |
| August 01, 2012 | 111.30 | 2833359.00 |

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